



# KEIRK IAN V. AVENIDO

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## OBJECTIVE

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To secure a position in a reputable organization where I can utilize my skills, problem-solving abilities, and academic background in Computer Engineering while gaining valuable industry experience and contributing to the company's success.

## SKILLS

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### Programming Languages:

- C++
- Python
- Java

### Tools & Frameworks:

- MATLAB
- Google Colab
- Canva

## WORK EXPERIENCE

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### Troubleshooting Engineer Intern at Sercomm Philippines Inc. – Calamba City, Laguna

April 6, 2026 – May 7, 2026

- Assisted in troubleshooting defective circuit boards and electronic devices.
- Performed rework and replacement of components such as ICs and shields.
- Supported technicians in testing and repair operations.
- Applied basic electronics, troubleshooting, and documentation skills in a production environment.

## EDUCATION

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### Bachelor of Science in Computer Engineering at National University -

Laguna 2022 - 2026

### Junior Highschool and Senior Highschool at Laguna College

2016 - 2022

### Elementary School at Angel's Kiddie Learning Center, Inc.

2010-2016

## CERTIFICATIONS

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- Computer Systems Servicing (NCII)
- Lean Six Sigma (Yellow Belt)
- Certified Safety Officer (SO2)
- Internship Completion Certificate – Sercomm Philippines Inc.

## PERSONAL BACKGROUND

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**Birthday:** June 09, 2004

**Citizenship:** Filipino

**Religion:** Catholic

**Civil Status:** Single

**Height:** 5'5

## PUBLICATIONS

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### YOLOv11-Based Pineapple Detection and Sorting System for Agricultural Automation

- **IEEE Published Conference Paper — 2026**

#### Co-Author

- Developed an automated pineapple grading and sorting system using YOLOv11, Raspberry Pi 5, and Arduino Mega.
- Contributed to image dataset preparation, model training, and system integration for agricultural automation.
- Published in IEEE conference proceedings for research in computer vision and embedded systems.

## PROJECTS

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### Predicting Student Performance Using Pandas and Machine Learning

- Developed a data-driven student performance prediction model using Python, leveraging Pandas for data cleaning, preprocessing, and exploratory analysis.
- Implemented and evaluated multiple machine learning algorithms (e.g., Logistic Regression, Random Forest) to identify key academic factors and generate performance predictions.
- Built a simple user-friendly interface/notebook workflow that allows users to input student data and receive instant performance predictions for academic decision-making.

### Digital Ears: Emotion Detection from Speech Signals in MATLAB

- Developed a MATLAB-based system that detects emotions (Happy, Sad, Angry) from speech using digital signal processing techniques such as normalization, framing, short-time energy (STE), and spectral analysis.
- Designed and implemented a full feature-extraction pipeline, generating structured datasets of frequency-based features for machine learning classification.
- Evaluated system performance using spectrograms and waveform analysis, ensuring accurate feature-label matching and consistent signal processing across varied audio samples.

### Computer Vision: Image Segmentation Techniques

- Built an image segmentation system using K-Means, Watershed, and edge detection methods, applying thresholding to achieve clear and accurate object outlines.

## TRAININGS AND SEMINARS

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- Open CV Bootcamp
- Prompt Engineering - Cognizant
- The Alpha of Data Science (Certificate of Completion)
- BOSH (Basic Occupational Safety and Health)

*I hereby certify that the above information is true and correct to the best of my knowledge and belief.*

KEIRK IAN V. AVENIDO  
Applicant's Signature